

Learning dynamical systems

First, what does this mean?

For discrete-time, autonomous system, this can be approximating $F: M \rightarrow M$.

Given: samples $x_1, x_2, \dots, x_{m+1}$ s.t.
 $x_t = F(x_{t-1})$

Think of this as supervised learning:

Given pairs $(x_t, F(x_t)) \quad t \in [m+1]$

$$\text{learn } \hat{F} = \min_{g \in \mathcal{G}} \sum_{t=1}^m \|g(x_t) - F(x_t)\|_{\mathcal{H}}^2$$

as $m \rightarrow \infty$, $\hat{F}$ minimum $L^2(\mu)$ interpolant of F in the class $\mathcal{G}$.

μ : typically natural invariant measure associated with F .

In stochastic systems, this can be applied to drift term, with known form of noise

Guarantees from statistical learning theory:

Generalization error / risk:

$$R(g) = \mathbb{E}_{x \sim \mu} \|F(x) - g(x)\|^2$$

Empirical error:

$$\hat{R}_m(g) = \frac{1}{m} \sum_{t \in m} \|F(x_t) - g(x_t)\|^2$$

This is what we actually minimize over a class of functions $\mathcal{G}$.

Naive application of learning bound gives:

For any $\delta \in (0, 1)$,

$$R(g) \leq \hat{R}_m(g) + 2 \text{Rad}(\mathcal{G}) + B \sqrt{\frac{\log 1/\delta}{2m}}$$

with Probability $\geq 1 - \delta$.

Here $l(x, g) = \|F(x) - g(x)\|^2$

So $R(g) = \mathbb{E}_{x \sim \mu} l(x, g)$

loss function

and $\hat{R}_m(g) = \frac{1}{m} \sum_{t \leq m} l(x_t, g)$

Probability space: The sample space is the set of "training examples". Here each element is the set $\{(x_t, F(x_t))\}$

$t \leq m$.

The distribution over this sample space is $(\mu \times \mu)^m$, assuming samples x_t are independent $\rightarrow$ this is usually not true when the data come from a single timeseries.

But it is a good approximation in chaotic / stochastic systems where there is a fast correlation decay.

$\text{Rad}(\log)$ refers to the
 "Rademacher complexity" of the
 function class $\log$
 comparison

There are many notions of function
 class complexity that are roughly
 speaking the one measure over
 the class ("size" of the class)

$$\hat{\text{Rad}}_m(F) = \mathbb{E}_{\sigma \sim \mathcal{R}^m} \left[\sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i f(x_i) \right]$$

set of scalar
 functions from
 some space $\mathcal{X}$ to
 $\mathbb{R}$

$S = (x_1, \dots, x_m)$
 $x_i \sim \mu$

$$\text{Rad}(F) = \mathbb{E}_{S \sim \mu^m} \hat{\text{Rad}}_m(F)$$

$$\sigma = [\sigma_1, \dots, \sigma_m] \text{ m iid}$$

σ : Rademacher random variables
 with distribution

$$\mathbb{P}(\sigma = 1) = \frac{1}{2}$$

$$\text{and } \mathbb{P}(\sigma = -1) = \frac{1}{2}$$

So, $\text{Rad}(F)$ measures how well functions f in F represent noise.

$$\left(\frac{1}{m} \sum_{i=1}^m \sigma_i f(x_i) \rightarrow \text{sample covariance between } \sigma \text{ \& } [f(x_1), \dots, f(x_m)] \right)$$

Usually G is a class of neural networks and $\text{Lo}G$ is the class of losses of neural networks.

Rad of this class is very large (vacuous) even when neural networks can be "compressed" (e.g. Arora Ge Neyshabur Zheng 2018)

So, this is not really informative about how ^{bound} well F is learned in the $L^2(\mu)$ sense

It is not designed to learn sequence-to-sequence maps (dynamics) or reproduce ergodic / perturbation-theoretic properties.